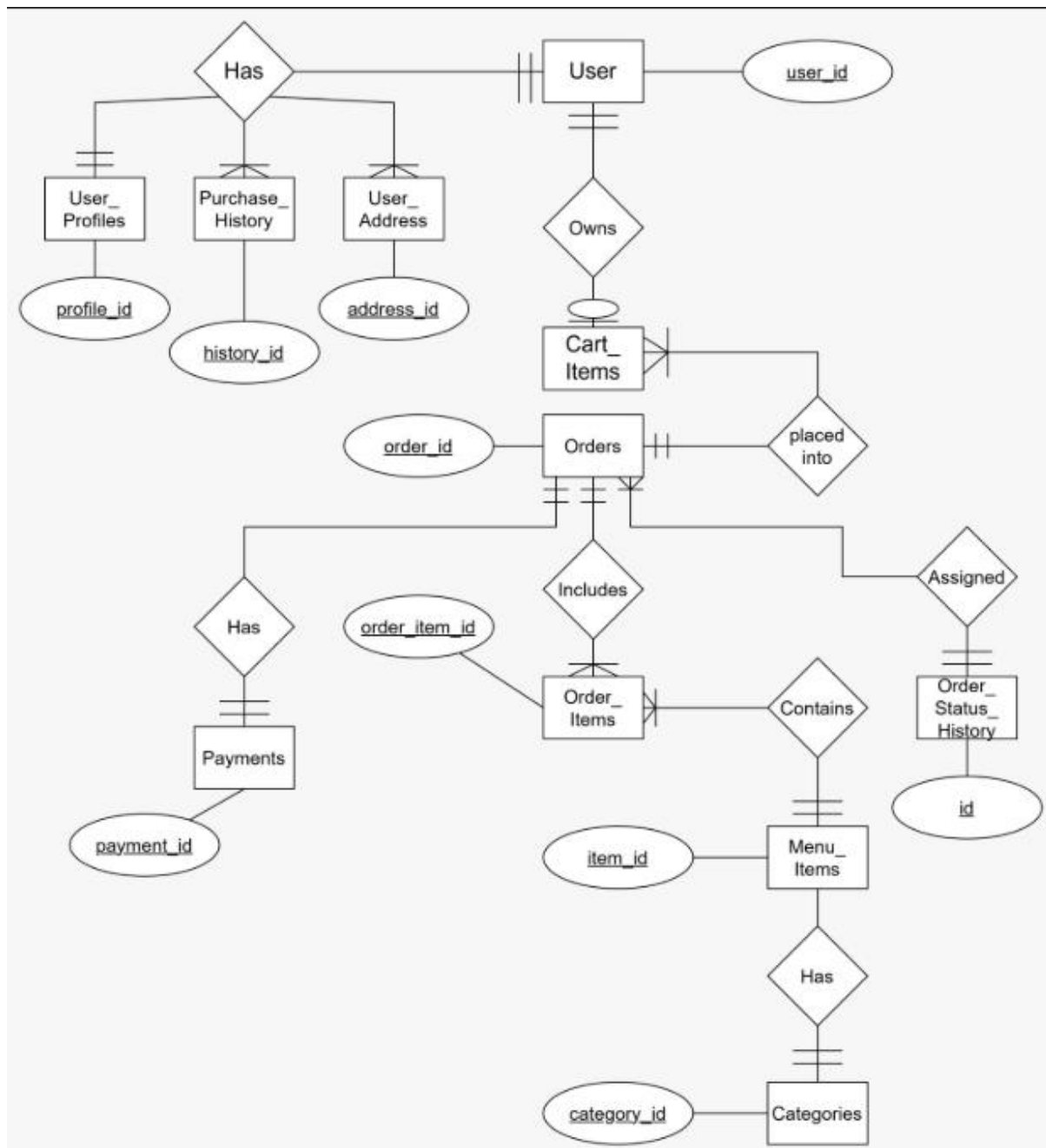


Name : Sharad Jhanwar(U24CS111)

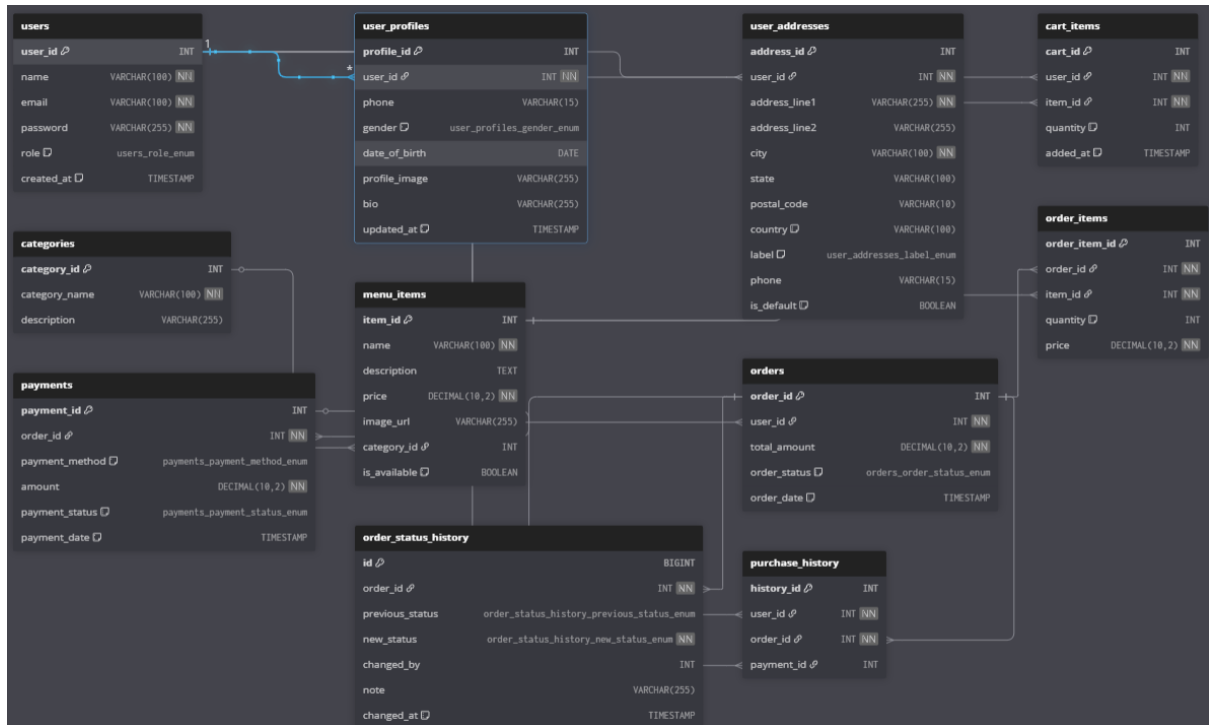
Name : Prathamkumar Vinodbhai Patel
(U24CS130)

Project Title: Zaika Restaurant

ER Diagram:



Database:



Entities and Table Structure

TABLE: USER

USER_ID(PK)

NAME

EMAIL(UNIQUE)

PASSWORD

ROLE

CREATED_AT

TABLE:USER_PROFILES

PROFILE_ID(PK)

USER_ID(FK)

PHONE

GENDER

DATE_OF_BIRTH

PROFILE_IMAGE

BIO

UPDATED_AT

TABLE:USER_ADDRESS

ADDRESS_ID(PK)

USER_ID(FK)

ADDRESS_LINE 1

ADDRESS_LINE 2

CITY

STATE

POSTAL_CODE

COUNTRY

LABEL

PHONE

IS_DEFAULT

TABLE:CATEGORIES

CATEGORY_ID(PK)

CATEGORY_NAME

DESCRIPTION

TABLE:MENU_ITEMS

ITEM_ID(PK)

NAME

DESCRIPTION

PRICE

IMAGE_URL

CATEGORY_ID(FK)

IS_AVAILABLE

TABLE:CART_ITEMS

CART_ID(PK)

USER_ID(FK)

ITEM-ID(FK)

QUANTITY

ADDED_AT

TABLE:ORDERS

ORDER_ID(PK)

USER_ID(FK)

TOTAL_AMOUNT

ORDER_STATUS

ORDER_DATE

TABLE:ORDER_ITEMS

ORDER_ITEM_ID(PK)

ORDER_ID(FK)

ITEM_ID(FK)

QUANTITY

PRICE

TABLE:PAYMENTS

PAYMENT_ID(PK)

ORDER_ID(FK)

PAYMENT_METHOD

AMOUNT

PAYMENT_STATUS

PAYMENT_DATE

TABLE:ORDER_STATUS_HISTORY

ID(PK)

ORDER_ID(FK)

PREVIOUS_STATUS

NEW_STATUS

CHANGED_BY

NOTE

CHANGED_AT

TABLE:PURCHASE_HISTORY

HISTORY_ID(PK)

USER_ID(FK)

ORDER_OD(FK)

PAYMENT_ID(FK)

Normalization:

Normalization is a process in database design used to organize data efficiently by removing redundancy (duplicate data) and ensuring data integrity. It divides large tables into smaller, related tables and defines relationships between them using keys.

Types of Normalization (Normal Forms):

1. 1NF (First Normal Form)

- Each column holds only **atomic (indivisible)** values.
- There are **no repeating groups or arrays**.
Example: Split multiple phone numbers into separate rows.

2. 2NF (Second Normal Form)

- Must be in **1NF**.
- All **non-key attributes** depend on the **entire primary key**, not part of it.
(Removes **partial dependency**.)

3. 3NF (Third Normal Form)

- Must be in **2NF**.
- No **transitive dependency** — non-key attributes should depend only on the key, not on other non-key attributes.

4. BCNF (Boyce–Codd Normal Form)

- A stronger version of 3NF.
- For every functional dependency ($X \rightarrow Y$), **X must be a super key**.
(Handles certain anomalies not covered by 3NF).

Use of Functions, Triggers, Procedures

The **functions**, **procedures**, and **triggers** together automate and organize operations in the database system efficiently:

- **Functions** are reusable routines that return specific calculated results, such as total customers, total revenue, average order value, or top-selling items. They perform quick lookups and summaries like monthly orders or revenue without needing repeated SQL queries.
- **Procedures** (like `GetUserFullProfile`) execute a series of SQL statements together — for example, retrieving all details of a user from multiple related tables (basic info, profile, and addresses) in a single call.
- **Triggers** automatically run when certain events occur. For instance, they update payment status when an order is completed, insert into purchase history when a payment succeeds, or remove deleted menu items from all user carts.

Together, these ensure automation, consistency, and efficiency in handling business logic within the database.